

John Abrahams

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EDUCATION

University of Pennsylvania

Expected December 2026

M.S.E. in Data Science · GPA 3.62/4.0

Coursework: Theory of Machine Learning, Emerging Theory of Large Language Models, Advanced Operating Systems

Lehigh University

May 2025

B.S. in Computer Science and Engineering · GPA 3.65/4.0 (major 3.75) · Minor: Probability and Statistics

EXPERIENCE

Ninth Wave, Software Engineer Intern, AI Team

May 2026 – Aug 2026

- Built a generator that sets up a new bank's AWS infrastructure from one short YAML file, replacing about ten hand edits per bank. Those hand edits had misrouted traffic for all three banks in production.
- Built Studio, an MCP gateway that gives AI tools one way into the company's internal apps under one set of access rules. Rewrote it in TypeScript after security review found two ways to pose as another user.
- Shipped eleven changes to the Spring Boot portal banks log into, including an activity timeline and an S3 file browser.
- Built pipelines that have Bedrock rewrite the docs weekly from the live AWS accounts (tested, not deployed).

New York Brain & Spine Surgery, Software Engineer

May 2025 – Mar 2026

- Built a therapy chatbot used in production on LLaMA and FastAPI: clinical rules steer each conversation, and the model writes the replies.
- Wrote the backend APIs and conversation-state handling, so prompts could be changed and tested quickly.

New York Brain & Spine Surgery, Research Associate

May 2023 – Aug 2023

- Co-authored a 236-patient study of outcomes after cervical spine fusion surgery, using logistic regression to find which factors predicted a fair or poor outcome.

RESEARCH

Reasoning and Memory in Language Models, independent research

Jul 2026 – Present

- Grafted a small recurrent reasoning module with learned halting (TRM-style) into the internal band a frozen language model uses for multi-step reasoning, training only about 5% of the parameters.
- Raised frozen Pythia-1B from near 0% to 54% accuracy on eight-hop questions, and to 61% on entities never seen in training; on Qwen2.5-Coder-7B the graft reached 96% on eight-hop questions.
- Now studying when a deployed language model should learn a new fact in its weights and when it should just look the fact up. Experiments on Pythia-410M and 1B with synthetic and real facts; not published.
- Found that looking facts up was as accurate as weight updates or better on every test, then priced the tradeoff: on Pythia-410M, learning a fact only saves compute once it has been asked about roughly 2,400 times.
- Built a gate that turns the update on only when a prompt names a learned entity, leaving all other text as it was.

PROJECTS

Kronos: a Type II exovisor, kronoslabs.dev

Oct 2024 – Present

- Built Kronos, a cloud for small software: a person or an AI agent deploys a Python or JavaScript program, and anyone with the link can run it without an account.
- Wrote the Rust runtime underneath: every program runs in its own WebAssembly sandbox with a time limit and can reach only the files and services its owner granted.
- Co-wrote a technical report with Ajax Li (not yet published) that audits every code path that starts a program and measures about 0.15 MiB of memory per added sandbox.

Crowd-Aware Music Recommender, UPenn ESE 5190, with Chris Spletzer

Aug 2025 – Dec 2025

- Wired ESP32 sensor boards and a thermal camera to a Raspberry Pi to track crowd movement and sound in a room.
- Trained a 1.8M-parameter GRU on about 3,000 scraped tracklists to predict the next song, plus a reinforcement learning head that adjusts each pick to how the crowd moves.

SKILLS

Languages: Python, Rust, TypeScript, Java, C++, Go

Machine Learning: PyTorch, LoRA fine-tuning, distillation, retrieval, reinforcement learning

Systems: WebAssembly, Wasmtime, AWS (CDK, Lambda, Bedrock, S3), Spring Boot, FastAPI, MCP, SQLite, Kubernetes (K3s), ESP32 and Raspberry Pi